DASHI: A Constructive Carrier Derivation Program for Physics-Unification Structures

Ultrametric Carrier Geometry, Projection-Defect Semantics, and Mechanized Closure Frontiers

DASHI project

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Abstract

Background: physics-unification programs often state target equations before making explicit which carrier, quotient, source, and calibration obligations are actually inhabited. Objective: this paper constructs a typed ultrametric carrier geometry for staging those obligations without collapsing theorem, target, obstruction, and empirical receipt into one status. Methods: DASHI combines a ternary/refinement carrier, `FactorVec` valuation coordinates, projection-defect decomposition, filtered operator surfaces, local transport, root-shell boundaries, UFT/motif compression, and mechanized admissibility receipts. Results: the manuscript derives a carrier spine from unresolvedness through recursive refinement, ultrametric distance, tracked lane commutation, projection residuals, and quotient-sensitive filtration; then applies that spine to gauge curvature, Yang-Mills ordering defects, QFT commutator targets, GR curvature/Bianchi targets, measurement residuals, sector splitting, compression admissibility, and a bounded below-Z Drell-Yan `t43` empirical contact surface (`chi2/dof = 2.1565191176` under threshold `4.0`). The same typed discipline blocks stronger readings: strict `chi2/dof <= 2` and strict log-covariance tests fail, upstream E8 completion is not promoted, and accepted empirical authority, sourced non-flat GR, continuum recovery, GRQFT closure, and completed unification remain open. Conclusion: DASHI proposes a constructive ultrametric carrier geometry in which admissible refinement, compression, transport, and quotient structure generate staged physics-facing derivation surfaces, while typed closure semantics track the residual obligations that block stronger promotion.

Keywords: ultrametric geometry; physics unification; gauge theory; filtered operator algebras; projection-defect decomposition; dependent type theory; constructive mathematics; formal methods.

1 Introduction

DASHI separates proved structure, target equations, empirical fits, and unresolved obstructions inside one constructive carrier geometry. A state carries trit-valued local data, tracked prime-lane valuations, ultrametric refinement depth, lane-local actions, and projections into observable surfaces. The residual left by a projection remains part of the formal state rather than being discarded or promoted by convention.

Observable structure is modeled as a projection from a richer carrier space. For a surface s, a projection $P_s : X \to Y_s$ exposes the admissible component of $x \in X$. The defect $D_s(x)$ records the residual structure: unresolved branches, incompatible quotient data, missing adapters, or absent empirical authority. Filtrations organize limiting behavior and associated graded constructions. Ultrametric refinement organizes address depth and precision. Independent lane actions give componentwise laws for selected coordinate sectors.

The same carrier supports local transport, filtered operators, ultrametric refinement, finite root-shell boundaries, compression structures, and bounded empirical receipts. Direction-indexed discrete sites define local transport schemas. Finite-support operators define filtered contraction targets. The tracked G6 commuting theorem gives a coordinate-independence law for selected prime-lane actions. Local E8/LILA surfaces record bounded root-geometry and upstream-promotion boundaries. UFT-style compression records repeated refinement motifs and recoverable residual distinctions. The bounded Drell-Yan empirical receipt records one comparison against a frozen typed surface.

Target Obligation Surfaces and DASHI Derivation Roles

The governing order for every displayed equation is

$$\text{carrier schema} < \text{interpretation map} < \text{field-equation law} < \text{calibrated physics claim}.$$

Until the interpretation map, quotient laws, physical units, source interface, and empirical/authority receipts are inhabited, the equations in this opening section are target obligation surfaces. They are placed here first because they are the reason the derivation spine matters, not because they are already closed. DASHI's current claim is that it gives a typed derivation route in which each term has an explicit carrier-side obligation:

$$\Omega_p = H_p - I, \qquad F_A = dA + A \wedge A, \qquad [\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma,$$

$$R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu},$$

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

The gauge and operator targets are similarly explicit:

$$d_A F_A = 0, \qquad d_A{*}F_A = J, \qquad dF = 0, \quad d{*}F = J,$$

$$U(t) = e^{-itH}, \qquad i\partial_t\psi = H\psi, \qquad [\phi(t,x), \pi(t,y)] = i\delta(x-y).$$

Physics term	DASHI structural reading	First residual obligation
$g_{\mu\nu}$	projected metric over refinement carrier	metric emergence, inverse, signature, units
∇	admissible transport law	non-flat connection/transport adapter
$R^\rho{}_{\sigma\mu\nu}$	noncommuting refinement transport defect	curvature carrier, covariance, torsion convention
$R_{\mu\nu}, R$	contracted transport defect	Ricci/scalar contraction and index laws
$G_{\mu\nu}$	stable curvature-obstruction tensor	contracted Bianchi/divergence-free law
$T_{\mu\nu}$	promoted source projection / matter residual carrier	stress-energy interface and authority
$\Lambda g_{\mu\nu}$	background refinement/vacuum term	cosmological convention and calibration
$8\pi G/c^4$	source-coupling normalization	weak-field/Newtonian limit and unit calibration

The normalization $\kappa = 8\pi G/c^4$ is not arbitrary decoration. In a completed route it must be fixed by geometric normalization, spherical-flux structure, and weak-field consistency with the Newtonian limit. Paper 1 does not prove that calibration. It makes the missing calibration a named obligation rather than hiding it behind the formal carrier.

Core Carrier Grammar

The constructions in this paper use a carrier of the form

$$x \in \mathcal{X} \simeq (a, \nu, \tau, d, \ell, r), \qquad \tau \in \{-1, 0, +1\}.$$

Here a is a refinement address, ν is `FactorVec`-style prime-lane valuation data, τ is the ternary coordinate, d is refinement depth, ℓ is the active lane, and r records residual/projection status. A surface s exposes only its admitted projection:

$$x = P_s(x) + D_s(x), \qquad \rho(x, y) = \alpha^{k(x,y)},\ 0 < \alpha < 1.$$

$P_s(x)$ is the promoted component; $D_s(x)$ is the residual defect; and $k(x, y)$ is the first incompatible address, valuation, or prefix depth. Filtered operator surfaces use $F_0 \subseteq F_1 \subseteq \cdots$, $\mathrm{gr}_n\, F = F_n/F_{n-1}$, and descended brackets only where quotient, equivalence, norm, and bracket laws are inhabited. Lane actions commute only on the official tracked independent `GL.FactorVec` route. UFT/motif compression records reusable projected structure while residual defects remain visible.

Derivation Map at a Glance

The closure frontier appears later. The early reader map is a derivation-role map: each target equation is read as a structural role to be produced by the carrier, followed by the adapter still needed for a calibrated physics claim.

Target role	DASHI derivation engine	Adapter still needed
Maxwell / abelian gauge	commuting local transports give additive plaquette curvature $F = dA$ with $dF = 0$ from $d_1 d_0 = 0$	bind adopted G2 schema into Maxwell scope; exterior derivative/nilpotency, Hodge star, source current, U(1) sector/action, covariance, units/convention
Yang-Mills / non-abelian gauge	noncommuting local transports force the commutator term $A \wedge A$ in $F_A = dA + A \wedge A$	representation, sector restriction, action/source law, calibrated field equation and coupling convention
QFT commutator	filtered finite-support operators descend only through quotient-correct brackets	equivalence modulo denominator filtration, accepted quotient carrier, descended norm/bracket, Hilbert/amplitude/CCR/Born laws
GR curvature	path-dependent carrier transport yields curvature as the minimal commutator obstruction	non-flat connection, Christoffel-from-metric law, Ricci/Bianchi reduction, stress-energy interface, units/calibration/source law
Measurement / uncertainty	projection $\pi : X \to Q$ leaves residual fibres $R_\pi(x)$	amplitude/probability law plus empirical projection and observable map
Sector splitting	projection preserves a subgroup $H \subseteq G$, leaving residual cosets G/H	concrete SM representation, subgroup projection law, coupling/mixing calibration

2 Background and Related Work

DASHI draws on standard mathematical neighborhoods including ultrametric and p-adic structure [18, 20], lattice and gauge language [14, 25, 11], contraction and associated graded constructions [23], root-system classification [10], mechanized mathematics [15, 19, 21], and compression-style trie geometry [17, 7]. The claims made here are restricted to the repo-local receipt and obstruction surfaces cited in the claim ledger.

Agda is used as the mechanized verification surface for selected propositions, receipts, audits, and blockers. A proved construction is represented by an inhabited artifact; a missing constructor is recorded as a manuscript-relevant obligation. This places the paper near proof assistants, dependent type theory, mechanized mathematics, and proof-carrying artifacts, while remaining narrower than an end-to-end certified physics pipeline.

Ultrametric and p-adic language organizes refinement by depth: nearby states share high-level valuation or prefix structure, while separation can occur at the first incompatible digit or coordinate. Hensel-style lifting is used only as an external analogy for compatible refinement across precision levels. In Paper 1 this is expository support for carrier and compression geometry, not an independent proof of convergence.

The labels G2, G3, and G6 are DASHI derivation-lane names. They should not be read as claims of equivalence to standard mathematical objects with similar names, such as the exceptional Lie group G_2, unless an inhabited surface states that equivalence. E8 is different: the paper explicitly uses standard root-system language there, but only for the bounded local enumeration and promotion-boundary surfaces stated below.

3 Derivation Spine

The carrier geometry is not introduced as an inventory. It is derived from a small sequence of representational obligations: unresolved local state, coordinate traversal, recursive refinement, prime-lane valuation, surface projection, residual defect, and filtered operator descent.

Stage	Object	Spine statement
1	Primitive ternary state	Signed evidence with unresolved residue requires at least three local statuses.
2	Traversal	Coordinate-local updates turn primitive states into paths in a finite state graph.
3	Voxel / hypercube cell	A finite block of ternary coordinates is a 3^k cell with coordinate-wall adjacency.
4	Trie and ultrametric	Recursive refinement is represented by prefix addresses whose natural distance is ultrametric.
5	`FactorVec`	Prime-lane valuation counts refine addresses into the tracked `Vec15 Nat` carrier.
6	Projection-defect	Every visible reading is a projection together with a residual defect.
7	Filtration	Depth, support, valuation, and admissibility organize surfaces into filtered theorem targets.

Definition 3.1 (Primitive local state)**.** Let $T = \{-1, 0, +1\}$. A primitive local state is an element $t \in T$. The values $+1$ and -1 record oriented alternatives. The value 0 records unresolved residue: structure that has not been discharged by the current surface.

Lemma 3.2 (Ternary minimality)**.** *Any local carrier that distinguishes a positive orientation, a negative orientation, and unresolved residue needs at least three local values.*

Proof. A two-valued carrier distinguishes at most two equivalence classes. If unresolved residue is identified with $+1$, unresolved structure is promoted as positive evidence. If it is identified with -1, unresolved structure is promoted as negative evidence. If $+1$ and -1 are identified, orientation is lost. The three roles therefore require three values. This is a bookkeeping minimality result, not a physical law. □

Definition 3.3 (Traversal)**.** For a finite coordinate set I, a primitive state is a function $x : I \to T$. A coordinate update is a map $u_{i,\phi} : T^I \to T^I$ that changes only coordinate i, using a local rule $\phi : T \to T$, and leaves every $j \neq i$ fixed. A traversal is a finite sequence

$$x_0 \xrightarrow{u_1} x_1 \xrightarrow{u_2} \cdots \xrightarrow{u_n} x_n.$$

Lemma 3.4 (Traversal as address word)**.** *Every traversal determines a word in the update alphabet $\Sigma = \{(i, \phi) \mid i \in I, \phi : T \to T\}$.*

Proof. Record the label (i_m, ϕ_m) of each update u_m. The traversal is represented by the word

$$(i_1, \phi_1)(i_2, \phi_2) \cdots (i_n, \phi_n) \in \Sigma^*.$$

The word is a formal update trace relative to the chosen carrier; it does not by itself claim a unique physical history. □

Definition 3.5 (Ternary cell and voxel refinement)**.** For a finite block $B \subseteq I$ with $|B| = k$, the local cell is $C_B = T^B$. It has 3^k vertices. Two vertices are coordinate-adjacent when they differ in exactly one coordinate. A voxel refinement replaces one coordinate i by a finite child block B_i, replacing the factor T at i by T^{B_i}.

Lemma 3.6 (Hypercube language is finite-coordinate language)**.** *The cube, voxel, and hypercube terms used in the origins story can be read as finite coordinate cells and their refinements.*

Proof. A block T^B is a finite product of local coordinate carriers. Its coordinate-adjacency graph is the ternary analogue of a cubical graph: faces are obtained by fixing coordinates, and refinements replace a coordinate by a child product. Thus the geometric language refers to finite-coordinate movement and refinement, not to an assertion that physical spacetime has been derived. □

Definition 3.7 (Refinement trie and ultrametric)**.** Let Σ be a finite refinement alphabet. A trie address is a finite or infinite word in Σ. For infinite addresses a, b, let $\lambda(a, b)$ be the length of their longest common prefix, with $\lambda(a, a) = \infty$. Fix $0 < \alpha < 1$, and define

$$\rho(a,b) = \begin{cases} 0, & a = b, \\ \alpha^{\lambda(a,b)}, & a \neq b. \end{cases}$$

Lemma 3.8 (Trie distance is ultrametric)**.** *The distance ρ satisfies*

$$\rho(a, c) \leq \max(\rho(a, b), \rho(b, c)).$$

Proof. If a and b agree for m symbols, and b and c agree for m symbols, then a and c also agree for m symbols. Hence $\lambda(a, c) \geq \min(\lambda(a, b), \lambda(b, c))$. Since $0 < \alpha < 1$, increasing prefix depth decreases distance, so the displayed ultrametric inequality follows. □

Definition 3.9 (`FactorVec`)**.** Let

$$P = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71\}$$

be the tracked supersingular-prime lane set used by the repo-native carrier. A `FactorVec` is a function $\nu : P \to \mathbb{N}$, equivalently a `Vec15 Nat` exponent vector with one coordinate for each tracked prime lane. For $p \in P$, define the prime-lane bump B_p by

$$B_p(\nu)(q) = \begin{cases} \nu(q) + 1, & q = p, \\ \nu(q), & q \neq p. \end{cases}$$

Lemma 3.10 (Coordinate bumps commute)**.** *For all $p, q \in P$, $B_p(B_q(\nu)) = B_q(B_p(\nu))$.*

Proof. Check each coordinate $r \in P$. If r is neither p nor q, both sides equal $\nu(r)$. If $r = p \neq q$, both sides equal $\nu(p) + 1$. If $r = q \neq p$, both sides equal $\nu(q) + 1$. If $p = q = r$, both sides equal $\nu(p) + 2$. The vectors are equal coordinatewise. □

Definition 3.11 (Surface projection and defect)**.** A surface s consists of a visible carrier Y_s, an admissibility gate G_s, and a projection $P_s : X \to Y_s$. The defect $D_s(x)$ is the residual information not promoted by P_s. When the carrier has an additive split, this is written $x = P_s(x) + D_s(x)$.

Lemma 3.12 (Defect makes non-closure visible)**.** *If two states $x, x' \in X$ satisfy $P_s(x) = P_s(x')$ but differ in gate status, residual content, or unmet obligations, then the projection alone cannot distinguish them. Recording $(P_s(x), D_s(x))$ keeps the promoted component and the unpromoted residual in the same formal account.*

Definition 3.13 (Filtration and associated graded target)**.** A filtration on X is an increasing sequence

$$F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots .$$

When quotient structure is available, the n-th graded piece is $\mathrm{gr}_n F = F_n/F_{n-1}$. This notation is a target unless the equivalence relation, quotient carrier, projection-respects-equivalence law, and descended operations are supplied.

Lemma 3.14 (Descent requires equivalence preservation)**.** *Suppose $A : X \to X$ is filtration-preserving and $x \sim_n y$ means $x - y \in F_{n-1}$. If $x \sim_n y \Rightarrow A(x) \sim_n A(y)$, then A induces an operator on F_n/F_{n-1}.*

Proof. Define the induced map by $[x] \mapsto [A(x)]$. If $[x] = [y]$, then $x \sim_n y$. By equivalence preservation, $A(x) \sim_n A(y)$, so $[A(x)] = [A(y)]$. Thus the definition is independent of representative. □

Theorem 3.15 (Early DASHI carrier spine)**.** *Starting from primitive signed state with unresolved residue, the carrier grammar is forced in the restricted sense*

$$\begin{aligned} &\textit{primitive state} \to \textit{traversal} \to \textit{voxel / hypercube cell} \\ &\qquad \to \textit{trie / ultrametric refinement} \to \texttt{FactorVec}\ \textit{valuation} \\ &\qquad \to \textit{projection-defect} \to \textit{filtration.} \end{aligned}$$

Proof. The lemmas above derive the route. Ternary minimality gives primitive local state. Traversal turns such states into coordinate-local update words. Ternary cells give the bounded reading of voxel and hypercube traversal. Recursive refinement gives prefix geometry and hence an ultrametric. Prime-lane refinement gives the `FactorVec` valuation route and basic coordinate commutation. Surface observation forces projection-defect accounting. Filtered operator language then requires quotient descent before theorem-facing dynamics can be promoted. □

This theorem does not assert that the grammar completes physics. It asserts that the listed structures are not a catalog: they arise as successive obligations once signed unresolved state, traversal, recursive refinement, prime-lane valuation, surface projection, and filtered operator descent are all kept visible.

4 Core Constructive Architecture

The carrier used below has six recurring components:

$$x \in X \simeq (a, \nu, \tau, d, \ell, r),$$

where a is a refinement address, ν is tracked prime-lane valuation data, τ is local trit state, d is depth or filtration grade, ℓ records the active lane/action context, and r records residual structure. The later gauge, operator, root-shell, compression, and empirical sections use different projections of this same carrier.

4.1 Carrier Coordinates and Depth

States are indexed by ternary local coordinates, refinement depth, and tracked prime-lane valuation vectors. Prime and supersingular-prime coordinates provide addressable lanes. Trit and Base369-style carriers distinguish positive, negative, and unresolved positions. A `FactorVec` assigns one valuation coordinate to each tracked prime lane, so lane updates can be checked componentwise.

Trie addresses induce an ultrametric: states are close when they share a long common refinement prefix, and they separate at the first incompatible depth. The carrier therefore records depth and precision without treating every refinement branch as an observed physical fact. Hensel-style lifting remains an external analogy for compatible partial information across increasing precision.

4.2 Projection-Defect Decomposition

The central local equation is the projection-defect split:

$$x = P(x) + D(x).$$

Here $P(x)$ denotes the promoted projection visible to a chosen surface, and $D(x)$ denotes the residual defect left by that projection. Projection exposes the admissible component of a carrier state while retaining the unresolved residual. Later promotion requires a typed bridge from $D(x)$ into the stronger surface.

4.3 Filtration, Lane Action, and Compression

Filtrations organize limiting behavior and associated graded constructions. For an operator surface, the key question is whether restricted operators descend to F_n/F_{n-1}, whether the quotient is well defined, and whether norms or brackets are preserved after projection. Lane actions are coordinate-local transformations on tracked sectors; the tracked G6 commuting theorem checks selected cross-lane actions on the current prime-valuation route.

Repeated local substructures permit refinement-address compression without collapsing admissible distinctions. UFT addresses record refinement structure, motifs record reusable projected substructure at a chosen grade, and residuals record distinctions that compression must preserve. A motif is not equality; it is evidence of reusable projection structure subject to the compression admissibility theorem below.

4.4 Closure Discipline

Closure in DASHI is a mathematical typing rule, not an editorial disclaimer. A surface promotes only the projection whose evidence is inhabited at that surface. The residual part remains visible as a defect: a missing quotient law, adapter, authority token, convention receipt, calibration, or upstream constructor.

This gives the manuscript one discipline for all lanes. A theorem surface may promote a local algebraic statement; an empirical receipt may promote a bounded comparison; a boundary record may promote only the fact that a stronger bridge is still absent. Diagnostics, analogies, provenance, and request surfaces can motivate later work, but they do not promote stronger physical claims.

```
surface S promotes projection P_s(x);
residual D_s(x) records the first missing bridge for stronger claim Q.
```

The intervening sections use this rule locally: each lane states its promoted surface, its first residual obligation, and no stronger reading unless the corresponding surface is inhabited.

5 Mathematical Constructions as Applications

The G2, G3, G6, and E8/LILA constructions instantiate the derivation spine on specific local carriers: discrete plaquette transport, finite-support operators, prime-lane commuting updates, and bounded root-shell geometry.

Label	Mathematical role	Current positive surface	Boundary
G2	Directed plaquette carrier schema	Direction-indexed SFGC schema	No Maxwell closure or field equation follows
G3	Finite-support operator filtration	Selected finite-support subtraction support	Associated-graded quotient descent remains open
G6	Prime-lane coordinate independence	Tracked cross-lane commuting theorem	Universal lane-operator promotion remains blocked
E8/LILA	Finite root-shell enumeration	Integer/two-sparse local completeness	Upstream E8 completion remains blocked

5.1 G2: Local Plaquettes from Coordinate Traversal

The local carrier is modeled by directed plaquette neighborhoods. A site has two visible movement directions; the minimal local cell records traversal in one coordinate, traversal in the other, and comparison around the resulting small square. This defines a checked interface for sites, directed edges, and plaquette-shaped neighborhoods.

The positive surface is the G2 schema-extension receipt together with the G2 no-remaining-missing witness. Direction-indexed SFGC schema adoption is complete for the current core surfaces. Dynamics remain open: curvature, field equations, continuum recovery, and calibration are deferred to the closure frontier. Exact Agda identifiers are listed in the reproducibility table.

5.2 G3: Selected Finite-Support Operators and the Quotient Gap

The G3 application starts where local traversal becomes operator calculus. The current constructive result is certified subtraction support for a selected constructive finite-support subtype, not arbitrary operator closure. Subtraction closure is required before filtered operators descend to stable quotient structures. The relevant inhabited surfaces are the selected finite-support subtraction support receipt, the selected subtraction support witness, and the selected subtraction certified-support record. A genuine associated graded layer still requires quotient classes of the form F_n/F_{n-1} and the kernel-equivalence law:

$$[x] = [0] \iff x \text{ lies in the denominator filtration piece.}$$

Until the accepted equivalence, quotient carrier, projection-respects-equivalence, and descended norm/bracket/isomorphism laws are inhabited, quotient notation is only a target shape. The current positive surface is selected finite-support subtraction support. The open obstruction is quotient-correct norm, product, bracket, and contraction structure.

5.3 G6: Distinct-Coordinate Commutation in Tracked `FactorVec`

The G6 lane records cross-lane independence inside the prime-coordinate carrier. An update to the `p` lane and an update to the `q` lane factor through distinct valuation exponents, so for `p != q` the order of those two tracked updates should not change the projected state. The tracked route proves that independent prime-lane updates commute componentwise. The positive surfaces are the tracked G6 commuting theorem and the tracked above-threshold coordinate-independence consumer. The old full `LaneOperator` interface is not promoted; its universal vanished-prime identity law blocks the nontrivial scaling route used by the tracked theorem. Migration from the tracked route to the older universal interface, or to downstream physics, remains uninhabited.

5.4 E8/LILA: Local Completeness and Upstream Promotion Boundary

The E8/LILA application uses the same spine in an enumeration setting. The carrier is finite local root geometry: integer and sparse local shapes can be enumerated, audited, and compared against the local semantic boundary. The positive and boundary surfaces are the E8 integer/two-sparse completeness theorem, the E8 upstream-promotion boundary, and the E8 upstream-promotion audit. The positive result is local integer/two-sparse completeness under the current audit boundary. The missing adapter is the upstream completion constructor or promotion API, not a stronger physics claim by local enumeration alone.

Lemma 5.1 (Root-shell emergence as bounded local geometry)**.** *Suppose a finite carrier region admits signed local coordinates, a norm or shell predicate, and an admissibility test that retains only locally compatible coordinate differences. Then the visible local states are organized first by shell membership and adjacency, not by arbitrary list position. When the admissible shell is finite, enumeration can certify completeness of that bounded shell and detect duplicate or missing local representatives.*

Proof. The norm/shell predicate partitions candidate local differences into finite fibres. The compatibility test removes candidates outside the current local geometry. Completeness is therefore a finite membership statement: every admissible representative in the shell has been listed, and no listed representative violates the predicate. This is the shape of the local E8/LILA root-boundary surface. It does not supply an upstream E8 completion token or a physical representation theorem. □

6 Locality, Time, and Causality from Finite-Support Traversal

Definition 6.1 (Finite-support configuration)**.** Let I be a coordinate index set and let $\mathcal{A}_i$ be the value space at coordinate i. A configuration is an admissible finite-support assignment

$$x \in \mathcal{C} \subseteq \prod_{i\in I}^{\text{fin}} \mathcal{A}_i, \qquad \operatorname{supp}(x) = \{i \in I \mid x(i) \neq 0_i\}.$$

Lemma 6.2 (Finite support gives a local boundary)**.** *If $x, y \in \mathcal{C}$ have finite support, then*

$$\Delta(x,y) = \{i \in I \mid x(i) \neq y(i)\}$$

is finite.

Proof. $\Delta(x,y) \subseteq \operatorname{supp}(x) \cup \operatorname{supp}(y)$, and the right side is finite. Thus any actual state difference has a finite coordinate witness. □

Definition 6.3 (Local move and traversal history)**.** A move $m : \mathcal{C} \rightharpoonup \mathcal{C}$ has finite footprint $\operatorname{foot}(m) \subset I$ when $m(x) = y$ implies

$$x|_{I\setminus\operatorname{foot}(m)} = y|_{I\setminus\operatorname{foot}(m)}.$$

A traversal history is a sequence

$$H = (x_0 \xrightarrow{m_0} x_1 \xrightarrow{m_1} x_2 \cdots)$$

with $x_{k+1} = m_k(x_k)$. Its internal time is the occurrence order of this sequence, not an assumed physical spacetime parameter.

Lemma 6.4 (Local moves preserve finite support)**.** *If x has finite support and m has finite footprint, then $m(x)$, when defined, has finite support.*

Proof. If $y = m(x)$, then

$$\mathrm{supp}(y) \subseteq \mathrm{supp}(x) \cup \mathrm{foot}(m).$$

The right side is finite. □

Definition 6.5 (Traversal filtration and causal preorder)**.** Define

$$S_n(H) = \bigcup_{k \le n} \mathrm{supp}(x_k) \cup \bigcup_{k<n} \mathrm{foot}(m_k).$$

Then $S_0(H) \subseteq S_1(H) \subseteq \cdots$. An event $e_k = (k, m_k, x_k, x_{k+1})$ directly depends on e_j when $j < k$ and the later move inspects a coordinate affected or certified by the earlier move, for example

$$\mathrm{foot}(m_j) \cap \mathrm{insp}(m_k) \neq \varnothing.$$

The reflexive-transitive closure gives an internal causal preorder subordinate to traversal order.

Lemma 6.6 (Finite-speed propagation in traversal steps)**.** *If the coordinate set carries a locally finite graph and every move footprint lies in a bounded neighborhood of its inspection set, then after n traversal steps the support of x_n lies in the n-step reachable region from the initial support.*

This is an internal locality theorem target. It does not yet assert relativistic locality, light cones, or spacetime completion. The residual obligations are the coordinate adjacency structure, admissible move class, finite-branching law, dependence predicate, and history-equivalence criterion.

Corollary 6.7 (Lorentz bridge as propagation-envelope target)**.** *If local moves have a uniform propagation bound and admissible histories preserve an invariant propagation envelope, then the reachability preorder determines cone-shaped regions*

$$\mathcal{C}^+(e) = \{e' \mid e \preceq e'\}, \qquad \mathcal{C}^-(e) = \{e' \mid e' \preceq e\}.$$

A Lorentz-like structure is the additional target in which these cones are represented by an invariant quadratic envelope, for example

$$q(\Delta) = c^2 \Delta t^2 - \|\Delta x\|^2 \geq 0,$$

up to the accepted signature and calibration laws.

Proof. Bounded traversal gives finite reachable regions at each step. If the bound is stable under admissible changes of local coordinates, then reachability cannot be represented as an arbitrary graph property; it becomes an invariant envelope of possible influence. The Lorentz target is the case where that envelope is coordinatized by a nondegenerate quadratic cone. The present paper does not inhabit that target; it names the adapter from finite traversal to invariant cone geometry. □

7 Gauge Transport from Local Sections

Definition 7.1 (Local section and transition)**.** Let C be the finite carrier and let $\pi : P \to C$ be a bundle target. A local chart over $U_i \subseteq C$ is a section

$$s_i : U_i \to P, \qquad \pi \circ s_i = \mathrm{id}_{U_i}.$$

On overlaps $U_{ij} = U_i \cap U_j$, a transition map is a gauge element

$$g_{ij} : U_{ij} \to G, \qquad s_j(x) = s_i(x) \cdot g_{ij}(x),$$

with target cocycle laws $g_{ii} = 1$, $g_{ji} = g_{ij}^{-1}$, and $g_{ij}g_{jk} = g_{ik}$ on triple overlaps.

The terminology in this section is the standard bundle, connection, and curvature terminology used in gauge geometry [14, 1, 12]; the DASHI contribution is the carrier-side staging of which pieces are inhabited, targeted, or still residual.

Definition 7.2 (Connection and plaquette defect)**.** A local connection assigns an oriented edge $e : x \to y$ a transport $U_e : F_x \to F_y$, or after trivialization an element $U_e \in G$. For a plaquette p with oriented boundary $e_1 + e_2 - e_3 - e_4$, define holonomy

$$\mathrm{Hol}(p) = U_{e_4}^{-1} U_{e_3}^{-1} U_{e_2} U_{e_1},$$

up to boundary convention, and discrete curvature defect

$$\Omega(p) = \mathrm{Hol}(p) - 1 \quad \text{or} \quad F(p) \approx \log(\mathrm{Hol}(p)).$$

Lemma 7.3 (Abelian transport reduces to a boundary sum)**.** *When link transports commute, plaquette holonomy can be represented additively by a cochain $a : E_1 \to A$:*

$$F(p) = d_1 a(p) = a(e_1) + a(e_2) - a(e_3) - a(e_4), \qquad a \mapsto a + d_0 \lambda.$$

Curvature is gauge invariant when $d_1 d_0 = 0$.

Definition 7.4 (Nonabelian target)**.** When transports do not commute, the order around the plaquette matters and the smooth target becomes

$$F = dA + A \wedge A, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu].$$

This is a target adapter, not a Maxwell or Yang-Mills closure. The residual obligations are carrier patches, local sections, transition maps, cocycle laws, oriented cells, finite incidence, $d_1 d_0 = 0$, a nondegenerate curvature carrier, and field-equation witnesses.

Corollary 7.5 (Why the abelian/nonabelian split is forced)**.** *If the transport group is abelian, the plaquette defect is a boundary-of-boundary object: $F = dA$, gauge change is $A \mapsto A + d\lambda$, and the homogeneous law $dF = 0$ follows from $d^2 = 0$. This is the DASHI route to an EM-shaped abelian sector. If the transport group is nonabelian, additive boundary calculus is not stable under composition; the commutator correction is the minimal local term that records the ordering defect, giving the Yang-Mills-shaped curvature $F_A = dA + A \wedge A$. The field equations $d{*}F = J$ and $d_A{*}F_A = J$ are not obtained from this split alone; they require a Hodge/source/action adapter.*

Theorem 7.6 (Plaquette ordering forces the Yang-Mills curvature form)**.** *Let $U_\mu(x) = \exp(\epsilon A_\mu(x))$ and $U_\nu(x) = \exp(\epsilon A_\nu(x))$ be infinitesimal transports in two local directions. The second-order plaquette holonomy has the form*

$$U_\nu(x)^{-1}U_\mu(x+\nu)^{-1}U_\nu(x+\mu)U_\mu(x) = I + \epsilon^2\big(\partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]\big) + O(\epsilon^3),$$

up to the chosen orientation convention. Thus the nonabelian curvature target is not an optional embellishment: the commutator term is the first order-sensitive local obstruction to plaquette closure.

Proof. Expand each edge transport to second order and compare the two orders around the plaquette. The derivative terms record the change of the local connection along neighboring edges. The only second-order term that survives solely because the two transports fail to commute is $[A_\mu, A_\nu]$. In the abelian case this term vanishes and the curvature reduces to dA. The residual obligations are the accepted smooth-limit adapter, representation, action/source law, Hodge operator, and calibrated field equation. □

8 QFT and Operator Surfaces from Filtered Quotients

Let A_{fs} denote the selected finite-support operator surface over the current G3 wave-function carrier W:

$$W = \mathrm{SelectedG3State} \to \mathbb{Q}, \qquad A_{\mathrm{fs}} \subseteq \mathrm{End}(W).$$

Let F_n be the selected filtration piece. The associated graded target is not F_n itself but

$$\mathrm{gr}_n(F) = F_n/F_{n-1}.$$

We use standard operator-algebra notation here [8, 16, 24]; the constructed object is the quotient-descent condition, not a field algebra.

Definition 8.1 (Quotient descent law)**.** The quotient projection $\pi_n : F_n \to \mathrm{gr}_n(F)$ is well formed only with an equivalence

$$A \sim_n B \quad\Longleftrightarrow\quad A - B \in F_{n-1}$$

and the kernel law

$$\pi_n(A) = \pi_n(B) \quad\Longleftrightarrow\quad A - B \in F_{n-1}.$$

Definition 8.2 (Descended bracket target)**.** The local bracket is

$$[A, B] = A \circ B - B \circ A.$$

The associated-graded bracket target is

$$[\pi_m(A), \pi_n(B)]_{\mathrm{gr}} = \pi_{m+n}([A, B]).$$

This definition is representative-independent only if

$$A \sim_m A',\ B \sim_n B' \Rightarrow [A, B] - [A', B'] \in F_{m+n-1}.$$

Definition 8.3 (Amplitude and commutator target)**.** A physical operator claim requires a structure map

$$\Phi : \mathrm{gr}(F) \to \mathrm{End}(\mathcal{H})$$

that preserves the descended bracket. A canonical commutator is therefore a residual target

$$R_{\mathrm{CCR}}(i, j) = [\Phi(X_i), \Phi(P_j)] - i\hbar\delta_{ij}I,$$

with promotion only after $R_{\mathrm{CCR}}(i, j) = 0$ is inhabited under an accepted state space, scalar/amplitude structure, identity, calibration, and quotient-descent receipt.

Lemma 8.4 (Why the commutator is the obstruction object)**.** *For operators that descend through π_n, the expression $A \circ B - B \circ A$ is the least local witness of order-dependence: it vanishes exactly when the two updates commute at the chosen projection grade. Thus the QFT-facing bracket does not enter as a decorative analogy. It is the obstruction to exchanging filtered updates after quotienting. The remaining physics step is the representation map Φ that turns that obstruction calculus into a calibrated operator algebra over $\mathcal{H}$.*

The residual obligations are the quotient carrier, kernel-equivalence law, finite-support product and commutator closure, bracket descent, amplitude space $\mathcal{H}$, bracket preservation, and physical calibration. QFT, local QFT, field algebra, Born rule, and interaction closure remain open.

9 Curvature and GR from Transport Defect

The GR target remains

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

In this paper the equation is a target: each term specifies a carrier-side construction still to be supplied. The left side requires metric, connection, curvature, Ricci contraction, scalar curvature, Einstein tensor, cosmological term, and a contracted-Bianchi law. The right side requires an authority-backed matter/stress-energy tensor, physical-unit normalization, and covariant conservation. The curvature and Bianchi notation follows standard GR conventions [22, 2, 13]; DASHI uses it here to name the exact adapter targets.

Definition 9.1 (Transport defect)**.** Let X be a discrete carrier with vertices x, oriented edges $e : x \to y$, and local fibres F_x. A local transport assigns $U_e : F_x \to F_y$. For a path $\gamma = e_n \cdots e_1$, let $U_\gamma = U_{e_n} \cdots U_{e_1}$. For a plaquette p based at x, define

$$H_p = U_{\partial p} : F_x \to F_x, \qquad \Omega_p = H_p - \mathrm{id}_{F_x}.$$

A non-flat curvature target requires a carrier where some plaquette defect is nonzero and transforms covariantly under frame change.

Definition 9.2 (Connection and curvature target)**.** A discrete connection can be treated as primitive transport $U_{x,\mu}$, or by an expansion

$$U_{x,\mu} = I + \epsilon\Gamma_\mu(x) + O(\epsilon^2).$$

The covariant difference target is

$$\nabla_\mu \phi(x) = U_{x,\mu}^{-1}\phi(x+\mu) - \phi(x).$$

For a GR route, this must supply $\Gamma^\rho_{\mu\nu}$, $g_{\mu\nu}$, $g^{\mu\nu}$, metric compatibility $\nabla_\lambda g_{\mu\nu} = 0$, and torsion-free symmetry where the Levi-Civita route is intended.

Curvature is the failure of covariant transport to commute:

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma - T^\lambda{}_{\mu\nu}\nabla_\lambda V^\rho.$$

The torsion-free GR target sets the torsion term to zero.

Definition 9.3 (Bianchi, contraction, and source)**.** The Bianchi targets are

$$R^\rho{}_{[\sigma\mu\nu]} = 0, \qquad \nabla_{[\lambda} R^\rho{}_{|\sigma|\mu\nu]} = 0.$$

Once $R^\rho{}_{\sigma\mu\nu}$ exists, the Ricci and scalar contractions are

$$R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}, \qquad R = g^{\mu\nu} R_{\mu\nu},$$

and the Einstein tensor is

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}.$$

The sourced equation then requires the same carrier, index convention, metric, connection, units, and physical regime for $G_{\mu\nu}$, $\Lambda g_{\mu\nu}$, and $T_{\mu\nu}$.

Lemma 9.4 (Why the Einstein tensor is the natural source-facing target)**.** *Once the Levi-Civita route is chosen, the contracted Bianchi identity gives*

$$\nabla^\mu G_{\mu\nu} = 0.$$

Any sourced curvature equation coupled to a conserved stress-energy target must therefore put a divergence-free curvature object on the left-hand side. The cosmological term is compatible because $\nabla^\mu g_{\mu\nu} = 0$. This does not prove the DASHI GR adapter; it explains why the adapter must target

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

rather than an arbitrary curvature expression. The value $\kappa = 8\pi G/c^4$ remains a weak-field/Newtonian calibration obligation.

The residual obligations are a non-flat connection, curvature covariance, finite-radius Bianchi identity, metric/inverse/contraction laws, contracted Bianchi law, physical stress-energy interface, unit constants G, c, Λ, and an inhabited sourced equation.

10 Measurement, Interference, Entropy, and Sector Splitting

Let X be a carrier state space and let

$$\pi : X \to Q$$

be a projection, coarse-graining, or quotient map into an observational state space. Two states are observationally equivalent when

$$x \sim_\pi x' \iff \pi(x) = \pi(x').$$

Measurement is modeled first as selecting or stabilizing a quotient representative in Q, rather than as direct access to x.

Definition 10.1 (Residual fibre)**.** The residual fibre of x is

$$R_\pi(x) = \pi^{-1}(\pi(x)).$$

Uncertainty is represented by the residual equivalence class left by the projection, pending a probability/amplitude law.

Theorem 10.2 (Incompatible projections force residual growth)**.** *Let $\pi_1 : X \to Q_1$ and $\pi_2 : X \to Q_2$ be two projections. If there is no projection $\pi_{12} : X \to Q_{12}$ and maps $r_i : Q_{12} \to Q_i$ such that*

$$\pi_i = r_i \circ \pi_{12} \quad (i = 1, 2),$$

while preserving the required distinctions of both Q_1 and Q_2, then the two observations admit no joint quotient representative. Sequential measurement must therefore leave an order-dependent residual

$$D_{12}(x) = \pi_2(s_1(\pi_1(x))) - \pi_1(s_2(\pi_2(x))) \,,$$

where s_i is any chosen section or representative rule. Nonzero D_{12} is the projection-defect route to uncertainty and interference targets.

Proof. A joint quotient representative is precisely a common refinement through which both projections factor. If no such refinement preserves the demanded distinctions, choosing a representative for one projection collapses distinctions needed by the other. The lost information moves into a residual fibre, and different section choices can produce different downstream projected states. A probability or amplitude law is still required before this becomes quantum measurement theory. □

Definition 10.3 (Interference and entropy targets)**.** If unresolved branches x_i combine before projection,

$$\Psi = \sum_i a_i x_i, \qquad \pi(\Psi)$$

can depend on branch combination before quotient separation. Entropy may be read projection-relatively as residual volume:

$$S_\pi(q) \sim \log \mu(\pi^{-1}(q)), \qquad S_\pi = -\sum_i p_i \log p_i$$

when branch weights p_i are supplied by the chosen formal model.

Definition 10.4 (Sector splitting)**.** Let $\mathcal{A}$ be an algebra of transformations on X. The projection-compatible subalgebra is

$$\mathcal{A}_\pi = \{a \in \mathcal{A} \mid \pi \circ a \text{ is well defined on } Q\}.$$

Commuting projected operations $[a, b]_\pi = 0$ form abelian sector targets. Noncommuting residual operations $[a, b]_\pi \neq 0$ point toward nonabelian targets. If a symmetry group G acts on X but projection preserves only $H \subseteq G$, then residual cosets G/H label directions no longer treated equivalently by the projected description. This is a target mechanism for sector splitting, not a derivation of the observed weak/strong/EM decomposition.

The residual obligations are the base category, the precise nature of π, branch-weight law, interference theorem, residual commutator, concrete group G, subgroup H, representation data, and empirical calibration.

11 Empirical Contact

The empirical result used in this paper is deliberately bounded: a below-Z Drell-Yan comparison against CMS-SMP-20-003 / HEPData `t43`. The receipt binds a named observable to a frozen comparison while leaving accepted authority, convention, and strict-fit obligations open.

The current positive surface is the bounded W3 below-Z Drell-Yan comparison-law receipt over CMS-SMP-20-003 / HEPData `ins2079374/t43`, using the `ins2079374/t44` covariance convention. The receipt records `chi2/dof = 2.1565191176`, threshold `4.0`, and mean prediction/data `0.9941233097`. It has no above-Z or accepted-authority promotion.

Two stricter protocols are explicitly negative. The adjacent-ratio receipt records that the stricter `chi2/dof <= 2` falsification-lane target is not satisfied. The strict log-covariance diagnostic is also negative: `phiStarRatioPredictor` has strict log `chi2/dof = 283.45739523864586`, and `sigmaDashiV4Predictor` has strict log `chi2/dof = 3180.211733150705`. The main obstruction is the log-linear residual subspace; the next theorem target is a frozen no-refit route discharging `StrictPassOrthogonalityObligation`, not a prose promotion.

Typed Residual Basis Decomposition

The empirical lane also exposes a reusable method: Typed Residual Basis Decomposition (TRBD). A TRBD receipt records

$$\begin{aligned}\text{residual} &\longrightarrow \text{projection onto a typed basis} \longrightarrow \text{named obstruction}\\ &\longrightarrow \text{coverage and perpendicular residual} \longrightarrow \text{promotion decision.}\end{aligned}$$

The reusable core is `DASHI.Core.TypedResidualBasisDecomposition`. The Drell-Yan strict-log receipt instantiates it with the structural basis $\{1, \log(\varphi^*)\}$. For `sigmaDashiV4Predictor`, the raw strict-log `chi2/dof` is `3180.211733150705`, the perpendicular `chi2/dof` is `111.96455543013676`, and the basis coverage is `0.968705212853035`. The coverage is high, but the perpendicular residual still fails the strict threshold `2.0`, so the TRBD status is `obstructionTypedPartial`; no promotable TRBD receipt is constructed.

TRBD distinguishes a typed obstruction from a passing shape law: identifying the residual subspace is useful, but it does not promote the predictor unless the complement is also discharged.

The bounded W3 comparison-law receipt is not accepted W3 empirical authority. No accepted provider token, non-postulated policy hook, or authority payload currently inhabits `W3AcceptedEvidenceAuthorityToken`. W4 and W5 remain blocked by missing accepted Drell-Yan luminosity/convention authority.

12 Compression, Semantic Geometry, and Cross-Domain Spine

UFT addresses compress repeated refinement structure. Two states may share a motif at grade Ω only when the distinctions required below Ω remain recoverable. A coarse address names a region of structure; descending the trie adds precision, distinguishes branches, and exposes incompatibilities. Branch distance is ultrametric, branch membership is projection-grade dependent, and unresolved branches can remain live without binary collapse.

Theorem 12.1 (Compression admissibility at a projection grade)**.** *Let Ξ be a semantic state, let Ω be the active projection grade, and let $\Pi_{\text{compress}}(\Xi)$ be a compressed representative. The compression is admissible at grade Ω only if every finer grade $\Omega' \leq \Omega$ has an explicit recoverability witness:*

$$\forall \Omega' \leq \Omega, \quad \text{RequiredDistinctions}(\Omega') \subseteq \text{Recoverable}_{\Omega'}\big(\Pi_{\text{compress}}(\Xi)\big).$$

The compression admissibility receipt inhabits this condition at motif grade for exact, PNF, and motif recoverability witnesses. It also records that the residual carrier is preserved, CID identity and semantic identity are separated, and false semantic collapse is excluded. The receipt status is diagnostic: it does not construct an external ITIR runtime authority token and does not identify byte-exact CID equality with semantic equivalence.

UFT-C, lattice-logic, null/hinge codes, SWAR implementation machinery, Base369/trit notation, Hensel-style refinement, and Markov-after-quotient readings are routed as background or appendix material unless a repo-local surface promotes a stronger claim.

The cross-domain variational-spine boundary records a common typed object $(X, \delta, \pi, \mathrm{defect}, \mathrm{gate}, \mathrm{observation}, \mathrm{symmetry})$ shared by several non-promoting target rows. Paper 1 does not infer chemistry closure, biological prediction, perceptual empirical fit, or universal closure from that boundary.

13 Closure Frontier

The closure frontier records the exact residual obligations remaining for stronger physical claims.

Gate	Current surface	First missing primitive/token	Unsupported stronger claim
G2	Direction-indexed SFGC schema adoption	Maxwell/curvature/field-equation carrier and receipts	Maxwell closure, physical curvature
G3	Selected finite-support subtraction support	Quotient carrier, kernel-equivalence law, descended laws	Global operator or Schrodinger closure
G6	Official tracked commuting route	Migration or split of old universal law	Old full `LaneOperator` promotion
E8/LILA	Local integer/root boundary	Upstream completion constructor/API	Upstream E8 completion
W3	Bounded Drell-Yan receipt	Accepted W3 evidence-authority token	Accepted empirical authority
W4/W5	Request/diagnostic surfaces	Accepted DY convention and correction receipts	Matter field, stress-energy, sourced physics
GR	Future adapter path	Non-flat connection, curvature, Ricci, sourced interface	Einstein tensor, Schwarzschild, continuum GR
Cross-domain spine	Shared projection-defect / MDL schema	Calibration, universality, domain receipts	Cross-domain recovery
Provenance	JMD/SSP/archive/source routing	Repo-local receipts or accepted authority	Provenance-as-proof

Every row identifies a residual defect: the exact bridge, authority, quotient law, adapter, or calibration object required for a stronger projection.

14 Reproducibility

The canonical manuscript, claim ledger, source ledger, archive-topic plan, and relative archive-location reference are included in the accompanying repository. Exact repository paths are listed in the supplemental repository materials rather than in the main prose.

The narrow verification surface for this draft is:

```
plantuml -checkonly Docs/*.puml
./scripts/render_docs_diagrams.sh
git diff --check -- Docs/PaperDraftWorkingFolder Docs README.md
```

Receipt Index

Agda identifiers are kept out of explanatory prose except where the exact name is the object under discussion. The table below gives the exact receipt names for the human-readable labels used in the body.

Receipt label	Module	Agda identifier
G2 schema extension	`DASHI.Physics.Closure.G2SFGCGaugeFieldSchemaExtension`	`canonicalG2SFGCGaugeFieldSchemaExtensionReceipt`
G2 no-missing witness	`DASHI.Physics.Closure.G2SFGCGaugeFieldSchemaExtension`	`noRemainingG2SFGCSchemaExtensionMissing`
G3 selected subtraction support	`DASHI.Physics.Closure.G3P2LimitConvergenceSurface`	`selectedFiniteSupportOperatorSubtractionCertifiedSupport`
G3 selected subtraction witness	`DASHI.Physics.Closure.G3P2LimitConvergenceSurface`	`selectedFiniteSupportOperatorSubtractionSupportWitnessFromCertified`
G3 certified-support record	`DASHI.Physics.Closure.G3P2LimitConvergenceSurface`	`G3SelectedSubtractionCertifiedSupport`
G6 tracked commuting theorem	`DASHI.Physics.Closure.G6CrossLaneCommutingTheorem`	`canonicalG6OfficialTrackedCrossLaneCommutingTheorem`
G6 above-threshold independence	`DASHI.Physics.Closure.G6AboveThresholdIndependence`	`canonicalG6OfficialTrackedAboveThresholdCoordinateIndependence`
E8 integer/two-sparse completeness	`DASHI.Algebra.Trit.E8RootEnumeration`	`integerIndexedRootsCompleteForTwoSparseShapeTheorem`
E8 upstream-promotion boundary	`DASHI.Algebra.Trit.E8RootEnumeration`	`canonicalE8RootEnumerationCompletePromotionBoundary`
E8 upstream-promotion audit	`DASHI.Physics.Closure.LilaE8RootEnumerationNoDuplicatesSurface`	`canonicalE8UpstreamCompleteReceiptPromotionAudit`
W3 bounded comparison law	`DASHI.Physics.Closure.HEPDataW3ComparisonLawReceipt`	`canonicalHEPDataW3ComparisonLawReceipt`
Drell-Yan adjacent-ratio lane	`DASHI.Physics.Closure.DrellYanAdjacentRatioEmpiricalLaneReceipt`	`canonicalDrellYanAdjacentRatioEmpiricalLaneReceipt`
Drell-Yan strict-log diagnostic	`DASHI.Physics.Closure.DrellYanStrictLogLinearSubspaceReceipt`	`canonicalDrellYanStrictLogLinearSubspaceReceipt`
Compression admissibility	`DASHI.Physics.Closure.CompressionAdmissibilityReceipt`	`canonicalCompressionAdmissibilityReceipt`
Cross-domain variational spine	`DASHI.Physics.Closure.CrossDomainVariationalSpine`	`canonicalCrossDomainVariationalSpineBoundary`

15 What This Paper Does Not Claim

This paper presents a constructive carrier derivation architecture and several partial inhabited surfaces. It does not claim completed empirical closure, sourced non-flat GR, GRQFT unification, upstream E8 completion, accepted W3/W4/W5 authority, or completed physics unification. Stronger readings remain conditioned on the residual obligations listed in the closure frontier.

Appendix A: Glossary and Naming Conventions

Term	Meaning in this paper
G2 lane	DASHI-native direction-indexed carrier schema for local plaquette/traversal constructions. It is not identified with the exceptional Lie group G_2 unless such an equivalence is explicitly inhabited.
G3 lane	DASHI-native filtered finite-support operator lane. The current positive result is selected finite-support subtraction support; associated graded quotient descent remains open.
G6 lane	DASHI-native tracked cross-lane commuting lane over `FactorVec` valuation coordinates. It is not a universal `LaneOperator` theorem.
E8/LILA	Local finite root-shell enumeration and audit surfaces associated with bounded E8-style root geometry. Upstream E8 completion is separate.
Surface	A projection context with a visible carrier, admissibility gate, projection map, and residual defect.
Inhabited	Constructed or proved in the current typed surface.
Target	A specified mathematical obligation shape that is not yet inhabited.
Receipt	A mechanized or repo-local witness record for a bounded claim.
Promotion	Admissible elevation of a claim to the status allowed by its evidence surface.
Residual / defect	Residual structure after projection.
Obstruction	A named missing bridge, law, token, adapter, or authority value that blocks a stronger claim.
Calibration	The units, constants, empirical convention, or source-coupling normalization needed for a physical claim.
Projection-defect decomposition	The reading $x = P_s(x) + D_s(x)$, where $P_s(x)$ is the admitted projection and $D_s(x)$ is residual structure.
UFT	Ultrametric Floating Trie: a hierarchical refinement-address structure where shared prefixes induce ultrametric distance.
`FactorVec`	Prime-lane valuation vector indexed by the tracked prime coordinate set.

Appendix B: Symbol and Object Index

Symbol	Meaning
$\mathcal{X}$	DASHI carrier state space.
x	Carrier state.
$(a, \nu, \tau, d, \ell, r)$	Address, valuation vector, ternary coordinate, refinement depth, active lane, and receipt/projection status.
$\tau \in \{-1, 0, +1\}$	Primitive ternary coordinate: opposed, unresolved, affirmed.
P_s	Projection admitted by surface s.
$D_s(x)$	Residual defect of x at surface s.
$\rho(x, y)$	Ultrametric distance between carrier states.
F_n	Filtration layer.
$\mathrm{gr}_n(F)$	Associated graded target F_n/F_{n-1}.
U_e	Transport operator on an oriented edge.
$\mathrm{Hol}(p)$	Plaquette holonomy.
$\Omega(p)$	Plaquette transport defect.
A, F_A	Connection one-form and nonabelian curvature target.
∇	Connection/covariant derivative target.
$R^{\rho}{}_{\sigma\mu\nu}$	Curvature target.
$R_{\mu\nu}, R$	Ricci and scalar curvature targets.
$G_{\mu\nu}$	Einstein tensor target.
$T_{\mu\nu}$	Stress-energy/source target.
$\pi : X \to Q$	Observation or quotient projection.
$R_\pi(x)$	Residual fibre $\pi^{-1}(\pi(x))$.
$H \subseteq G$	Preserved subgroup under sector projection.
G/H	Residual coset sector.

Appendix C: Origins from Ternary Traversal to Carrier Geometry

The earliest DASHI experiments used cyclic ternary and nonary carriers. A trit distinguished positive, negative, and unresolved states; larger wrapped state spaces represented transition, tension, overflow, and refinement without collapsing the unresolved case into a binary decision. Later dialectical language in the archive corresponds to the same mechanism: a transition may carry unresolved residue, and the current projection records that residue as $D_s(x)$.

The geometric intuition then moves from a one-dimensional cycle to traversal through a cube-like state space. Rubik-like cube, hypercube, and voxel language is useful only as intuition for finite-coordinate movement: each coordinate records a local trit, phase, or valuation, and a traversal is a sequence of admissible coordinate updates. Voxel language fits the same role at nonary scale: a local 3^2 or 3^k block can be read as a small cell of structured state, and overflow means that the current cell cannot preserve the relevant invariant without adding a refinement coordinate.

Early implementations tested traversal continuity using sampled video/state traces. A video trace is a sampled path through state space: continuity testing asks whether adjacent frames preserve the expected local carrier relation, whether an apparent jump is an admissible carry/refinement, and whether the projection defect increases at the boundary. Such tests motivated carrier choices and exposed discontinuities. The experiments are heuristic unless bound to a reproducible metric and verification surface.

The sibling inventories record this boundary. Early implementations explored ultrametric refinement, traversal continuity, diffusion rollout, branch-density localization, and adaptive refinement. One tree-diffusion diagnostic records `tree_rollout_mse = 4.246448141002339e-06` versus `rbf_rollout_mse = 2.155684319973986e-05`. These experiments motivated the refinement and transport structures later formalized in the carrier; they are not theorem-level evidence.

The selected figures are copied into `Docs/Images/paper1-origin-figures/` and recorded in `Docs/PaperDraftWorkingFolder/FigureCandidateManifest.md`. They illustrate the refinement and traversal geometries that motivated the carrier formalism.

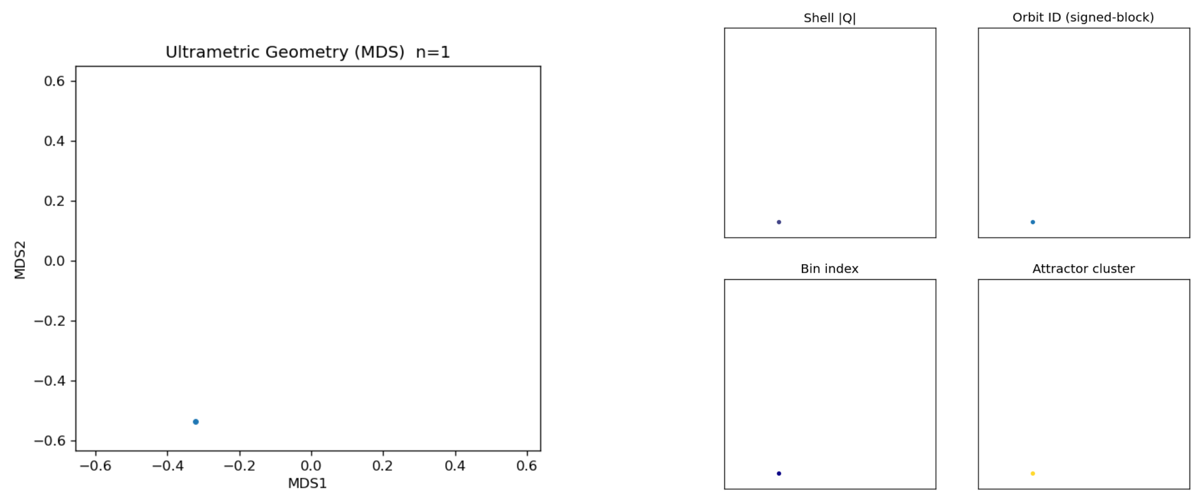

Figure A.1. Ultrametric refinement geometry. States sharing longer refinement prefixes cluster naturally under the induced ultrametric; divergence occurs at the first incompatible address or valuation depth.

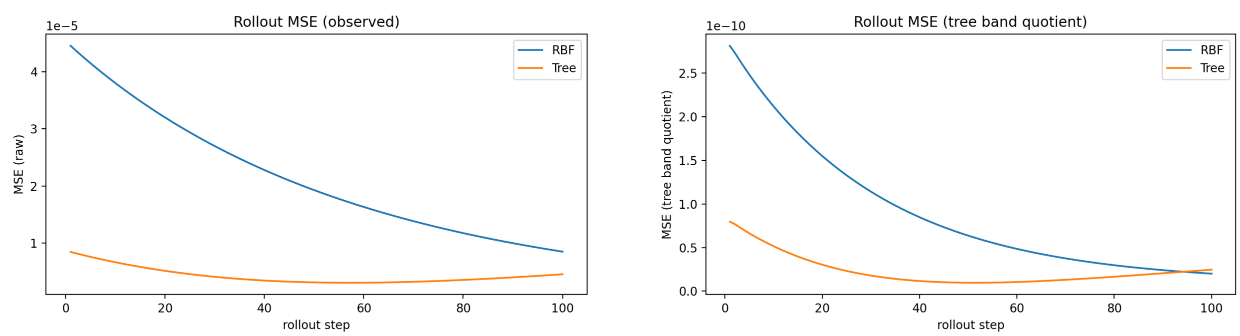

Figure A.2. Traversal continuity and tree-diffusion rollout diagnostics. The sibling run gives a concrete visual and metric surface for tree-structured transport; it does not promote a video-continuity or codec-correctness theorem.

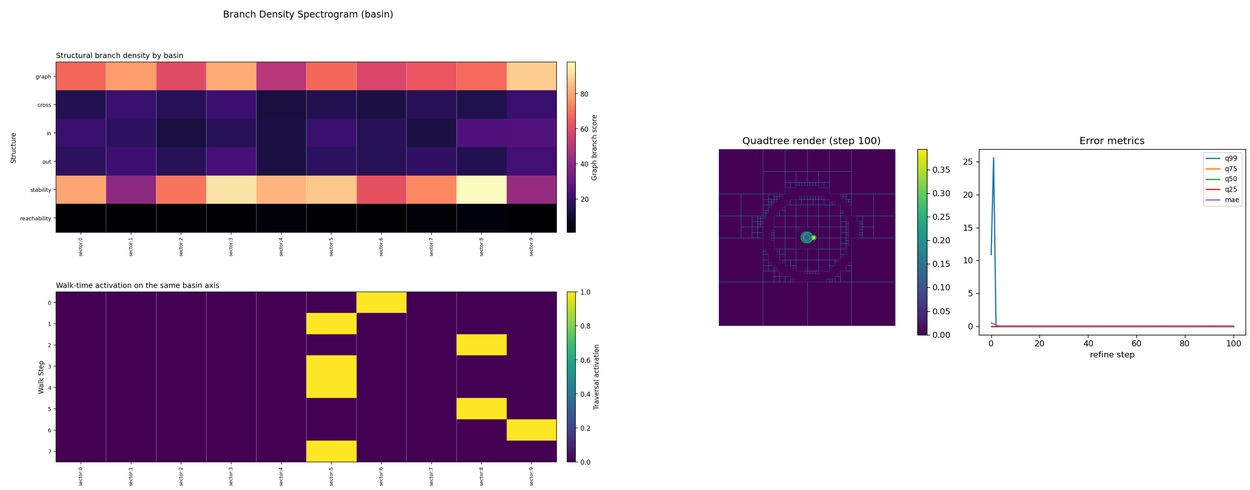

Figure A.3. Branch density and adaptive refinement. The diagnostics show refinement concentrating around structured local regions rather than spreading uniformly across the carrier.

UFT/trie refinement gives this traversal a compression geometry. A trie node is a coarse address; descending the trie adds precision. Nearby branches share prefix structure, and separation occurs at the first incompatible depth. This matches the ternary ultrametric reading already used in the carrier spine: refinement depth is not just a storage index but a metric and semantic relation. A motif is then a reusable local shape in the trie, not an equality claim.

`FactorVec` is the corresponding prime-coordinate refinement of the same idea. Instead of only trit positions, a state carries valuation data across tracked prime lanes. Multi-lane transport, chamber legality, signed scans, and defect summaries turn traversal into a controlled carrier geometry: a move may be legal or illegal, stable or repatterning, contractive or expansive, and these labels are computed at the representation layer rather than inferred from prose.

The transition from trits to carrier geometry is therefore historical and mathematical. Ternary traversal made unresolvedness explicit; `Base369`, cube/voxel traversal, video-like continuity traces, UFT tries, and `FactorVec` lanes added progressively sharper ways to represent transition, address, refinement, and defect. These origins do not prove full physics unification, automatic convergence, Monster representation semantics, empirical adequacy, or domain recovery. They record the path by which stronger claims would have to be typed, projected, tested, and closed.

16 Conclusions

DASHI is a constructive physics-unification program centered on a coherent carrier geometry. Prime-coordinate structure, ultrametric refinement, projection-defect decomposition, filtered operator surfaces, tracked lane actions, local root-geometry boundaries, semantic compression, and bounded empirical receipts are treated as parts of one mathematical construction.

The current frontier is substantial but not final. G2 records a direction-indexed carrier schema, G3 records selected finite-support subtraction support, G6 records tracked cross-lane commutation, and E8/LILA records local root-boundary structure. UFT and motif structures keep refinement, semantic distance, and residual structure in the same typed vocabulary; the Drell-Yan receipt records bounded empirical contact and a stricter residual shape law that still fails.

The residual obligations remain explicit throughout the construction, so stronger physical claims require the corresponding adapters before promotion. Paper 1's contribution is a constructive ultrametric carrier geometry whose refinement, transport, projection, compression, and quotient structures support staged physics-facing derivation surfaces with a visible certification frontier.

References

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